REVERSE ENGINEERING – ACTIVITY REPORT

PART I – STUDYING THE PRODUCT

1. External Features

Physical Construct

The device features an ergonomic, lightweight black plastic shell composed of an upper top housing and a bottom structural base. It is designed with a symmetrical profile accommodating both right- and left-handed palm grips. Primary user controls include standard primary buttons (left and right click), a central textured scroll wheel, and an integrated rubberized cable strain relief at the front lead.

Subsystems Used

- **Mechanical Interface:** Outer protective casing, tactile microswitches, scroll wheel assembly, and mechanical rotary encoder.
- **Optical Sensing Subsystem:** IR/Red LED light source, optical sensor Integrated Circuit (IC), and plastic light-focusing lens aperture.
- **Microcontroller & Interface Subsystem:** Main optical IC/controller and integrated 4-wire USB interface wiring harness.
- **Power Regulation & Decoupling Subsystem:** Passive resistors and decoupling capacitors for power line smoothing and signal biasing.

Description of Subsystems

The outer enclosure acts as a rigid mechanical housing designed to retain and protect a single main single-sided Printed Circuit Board (PCB). The mouse registers discrete click events through mechanical tactile microswitches and continuously tracks two-dimensional surface movement via an optical sensor chip operating alongside a focused illumination source.

Modes of Operation

- **Plug-and-Play USB Wired Mode:** Operates under standard continuous 2D motion tracking and Human Interface Device (HID) signal generation, powered directly by a 5V DC supply delivered over the host USB bus.

2. Purpose of the Product

Provides precise two-dimensional cursor motion tracking, scroll navigation, and digital user-input signal feedback for desktop and laptop computing environments.

3. User Needs Addressed

- **High Reliability & Precision:** Low-latency continuous cursor tracking operating without the need for physical mechanical rollers or tracking balls.
- **Ergonomic Design:** Curved outer profile providing comfortable support during extended computing sessions.
- **Plug-and-Play Simplicity:** Instant deployment without requiring battery maintenance, external charging, or wireless pairing procedures.

PART II – PRODUCT ORIGIN TO LATEST INVENTIONS

1. Difference in Technology Across Versions

1.1 Early Stage (Mechanical Trackball Mice)

Utilized a rubber-coated heavy steel ball driving two orthogonal mechanical rollers connected to slotted rotary encoder wheels. These systems were highly susceptible to dust, lint, and oil accumulation, requiring frequent manual disassembly and cleaning.

1.2 First Generation (Early Optical Mice)

Replaced physical tracking balls with an LED and low-resolution optical sensors. These early models required specialized grid-patterned metallic mouse pads to detect movement vector changes.

1.3 Second Generation (Standard LED Optical Mice)

Introduced integrated high-speed optical surface imaging processors combined with Digital Signal Processors (DSPs). This allowed real-time pattern tracking on arbitrary non-reflective surfaces without dedicated mouse pads.

1.4 Latest Generation (Laser & High-DPI Wireless Sensors)

Utilizes invisible laser diodes or ultra-high-DPI optical sensors capable of tracking across polished or reflective glass surfaces. Modern iterations incorporate ultra-low-power wireless protocols (Bluetooth / 2.4GHz) and dynamic power management.

2. Current Status – Core Smart Technologies Used in Product

The unit under analysis utilizes an integrated single-chip optical mouse sensor (U1). This package combines optical image acquisition, motion vector processing algorithms, and USB protocol transmission directly on a single silicon die.

3. Engineering Importance of This Evolution

- **Extended Service Life:** Complete elimination of mechanical moving parts for primary movement tracking significantly reduces component wear and extends operational lifespan.
- **System Optimization:** System-on-Chip (SoC) integration drastically reduces total PCB component count, surface area requirements, manufacturing assembly steps, and overall product bill-of-materials (BOM) cost.
- **Performance Enhancement:** Delivers higher tracking resolution (DPI), lower signal response latency, and minimal energy consumption.

PART III – INTERNAL COMPONENTS & WORKING FLOW

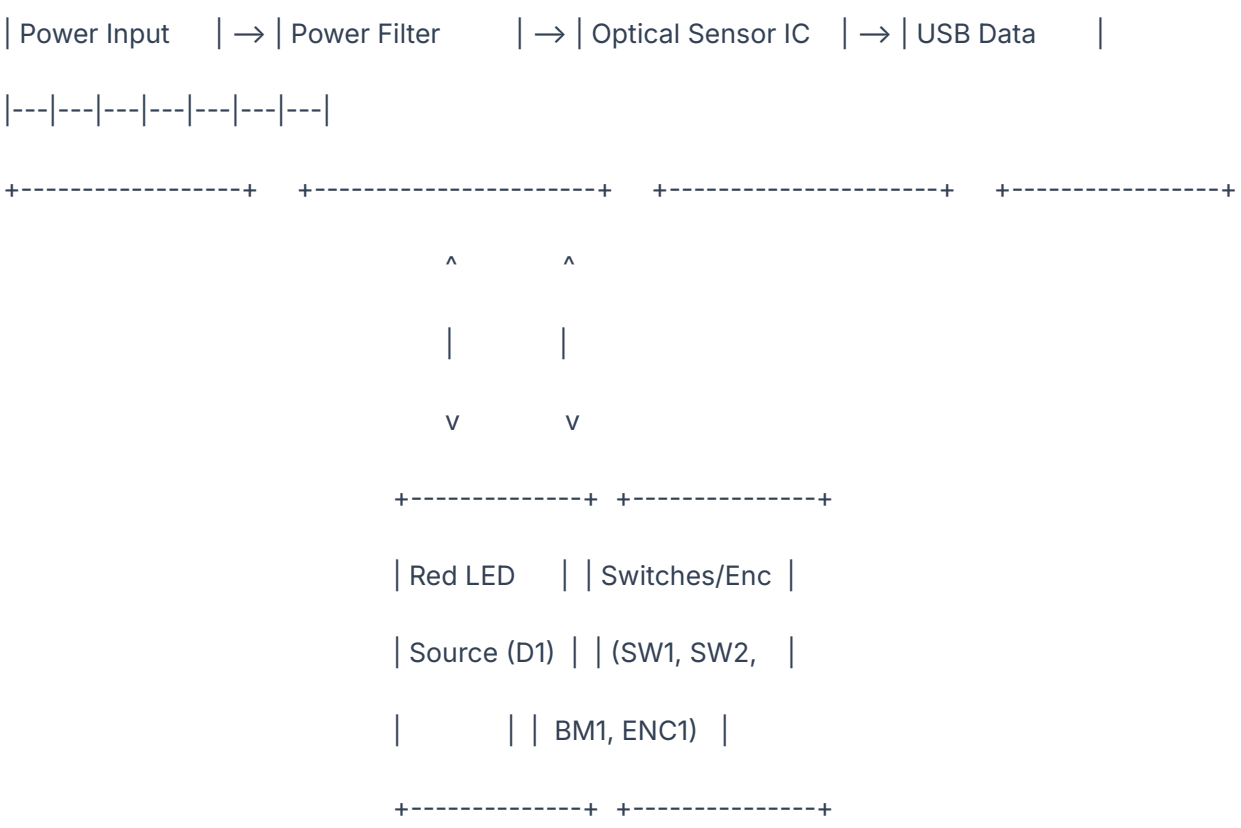
1. Introduction of the Product

This tear-down report analyzes a standard Dell wired USB optical mouse. The hardware structure relies on a single-sided paper-phenolic (FR-2) printed circuit board identified by board markings 812-00954-01A and **SM8122_UB**.

2. Photograph of the Internal Components of the Product

Figure 1a: Top-down view of PCB inside housing showing scroll wheel, switches, and DIP-8 optical IC.
Figure 1b: Bottom enclosure view showing model label CN-0X9DCG-71616-529-0LGH and optical lens opening.
Figure 1c: Close-up view of bare PCB substrate showing component annotations SM8122_UB, 812-00954-01A, resistors, and capacitors.

3. Internal Block Diagram (Functional View)



4. Internal Subsystems with Detailed Explanation

4.1 Function of the Subsystem

- **Optical Sensing Subsystem:** Handles surface illumination and continuous microscopic image capture to process directional displacement.
- **Mechanical Switch Assembly:** Captures discrete user click inputs for primary and secondary actions.
- **Scroll Encoder Assembly:** Detects rotary position increments and direction during wheel rotation.

4.2 Working Principle

The primary LED (D1) casts angled light onto the underlying surface through a molded optical lens. The reflected light patterns are focused back onto a CMOS sensor array inside U1. An integrated DSP compares consecutive image frames at high frequencies to calculate relative X-axis and Y-axis displacement vectors.

4.3 Characteristics

- **Operating Voltage:** 5V DC supply via standard USB bus wiring.
- **Interface Standard:** USB Human Interface Device (HID) protocol.

- **PCB Construction:** Single-sided FR-2 phenolic substrate utilizing top-side jumper wires for single-layer routing.

4.4 Role / Engineering Justification

Deploying a single-chip sensor and controller eliminates the need for discrete microcontrollers, external clock oscillators, and separate optical processing hardware, minimizing unit cost and surface area footprint.

5. Complete Working Flow (Step-By-Step)

1. **Power Delivery:** Plugging the USB cable into the host system provides a 5V DC supply to the PCB power rails.
2. **Surface Illumination:** Resistor-regulated current drives LED D1, illuminating the surface beneath the optical aperture.
3. **Image Acquisition:** The optical array inside U1 continuously captures microscopic surface profiles at multi-kilohertz frame rates.
4. **Vector Calculation:** The internal DSP executes surface correlation algorithms to calculate incremental delta X (ΔX) and delta Y (ΔY) movements.
5. **State Detection:** Button actuation or scroll wheel rotation shifts the digital logic states on the input pins of U1.
6. **Data Transmission:** IC U1 packages the movement vectors and switch states into standardized HID data packets and transmits them over USB data lines (D+, D-) to the host system.

6. Components List – Bill of Materials (BOM)

Ref. Des.	Component Name / Description	Package / Type	Qty
U1	Optical Mouse Sensor & Controller IC	DIP-8 Optical Package	1
SW1, SW2	Left / Right Tactile Microswitches (TTC)	3-Pin Through-Hole	2
BM1	Scroll Wheel Button Switch	2-Pin Sub-miniature	1
ENC1	Rotary Mechanical Scroll Encoder	Through-Hole 8mm Height	1
D1	Red / IR Illumination LED	5mm Clear Through-Hole	1

Ref. Des.	Component Name / Description	Package / Type	Qty
C4, C5	Power Decoupling Capacitors (Ceramic & Electrolytic)	Radial Lead Through-Hole	2
R1–R8	Carbon Film Resistors	Axial Through-Hole	8
J1	4-Pin USB Cable Header Connector	4-Pin Straight Header	1

7. Component Ratings & Specifications

Component	Parameter / Marking	Value / Specification
PCB Substrate	Markings: `812-00954-01A` / `SM8122_UB`	Single-Sided Phenolic FR-2, 1.6mm thickness
Optical IC (U1)	8-Pin Optical DIP Package	Operating Voltage: 5.0V DC; Integrated DSP & USB PHY
Switches (SW1/SW2)	TTC Blue Plunger Switches	Rated: 5V DC, 50mA; Expected Life: 1,000,000+ Cycles
Scroll Height (ENC1)	Board Marking: `JP1 = 8mm`	8mm Center Axis Height
USB Cable	4-Wire Shielded Assembly	Standard 5V USB 2.0 (VCC, GND, D+, D-)

8. Component Function Table

Component Name	Function	Specifications
Optical Sensor IC (U1)	Captures surface reflection images, calculates directional movement vectors, reads switch logic, and manages USB communications.	Integrated 8-pin optical DIP package with built-in sensor array.
Illumination LED (D1)	Casts steady illumination onto the surface below the sensor lens.	5mm transparent casing, current-limited via series resistor.

Component Name	Function	Specifications
TTC Switches (SW1/SW2)	Translates mechanical finger pressure into standard binary click signals.	Momentary SPST switch mechanism.
Mechanical Encoder (ENC1)	Measures rotational steps and direction of the scroll wheel.	Dual-channel mechanical rotary encoder, 8mm axis profile.
Electrolytic Cap (C5)	Smooths low-frequency voltage ripples on the 5V DC incoming power line.	Polarized radial aluminum capacitor.
Axial Resistors (R1–R8)	Limit current to the primary LED and provide pull-up/pull-down bias for signal lines.	Standard carbon film resistors with axial leads.